ARTIFICIAL INTELLIGENCE INTERNSHIP

AI Internship at CodSoft

Chatbot with Rule-Based Responses • Tic-Tac-Toe AI • Recommendation System

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What We'll Cover

01

About CodSoft

Organization profile & internship program

02

Internship Overview

Domain, duration, and objectives

03

Task 1 — Chatbot

Rule-based conversational agent

04

Task 2 — Tic-Tac-Toe AI

Minimax with Alpha-Beta Pruning

05

Task 3 — Recommendation System

Collaborative & content-based filtering

06

Tools, Outcomes & Certificate

Technologies used and key takeaways

CodSoft

Technology internship & training platform

About the Organization

- Provides hands-on, project-based internship programs to students
- Domains: Artificial Intelligence, Web Development, Android Development, Python & Java Programming, Data Science
- Learning model: structured task sheets + independent implementation + mentorship
- Focus on building industry-relevant, practical problem-solving skills

4 Weeks

Virtual Internship in Artificial Intelligence | 15 Jul 2026 – 15 Aug 2026

Internship Overview

The internship focused on building a strong practical foundation in Artificial Intelligence through three independent mini-projects, each targeting a core AI concept — conversational logic, adversarial search, and personalization.



Conversational AI

Rule-based logic & pattern matching for natural dialogue flow



Game-Playing AI

Adversarial search using Minimax & Alpha-Beta Pruning



Personalization

Collaborative & content-based recommendation techniques

TASK 1

1

Chatbot with Rule-Based Responses

A simple chatbot that responds to user inputs using predefined rules — an introduction to conversation flow and basic NLP.

1. Used if-else statements and pattern matching to identify user queries
2. Mapped recognized query patterns to appropriate predefined responses
3. Built understanding of Natural Language Processing fundamentals and dialogue design

TASK 2

2

Tic-Tac-Toe AI

An unbeatable AI agent that plays Tic-Tac-Toe against a human opponent using classic game-theory search.

1. Represented the 3x3 board as a matrix and modeled all possible game states
2. Implemented the Minimax algorithm to evaluate optimal moves recursively
3. Applied Alpha-Beta Pruning to optimize search and speed up decision-making

TASK 3

3

Recommendation System

A simple system that suggests movies, books, or products to users based on their preferences.

1. Explored collaborative filtering — recommending items liked by similar users
2. Explored content-based filtering — recommending items similar in attributes
3. Tested both approaches on a small sample dataset of user preferences

Technologies & Tools Used

Python 3

Core programming language for all three tasks

Minimax & Alpha-Beta Pruning

Game theory & adversarial search algorithms

Pattern Matching / Regex

Rule-based logic for the chatbot

Pandas / NumPy

Data handling for the recommendation engine

VS Code / Jupyter Notebook

Development environment

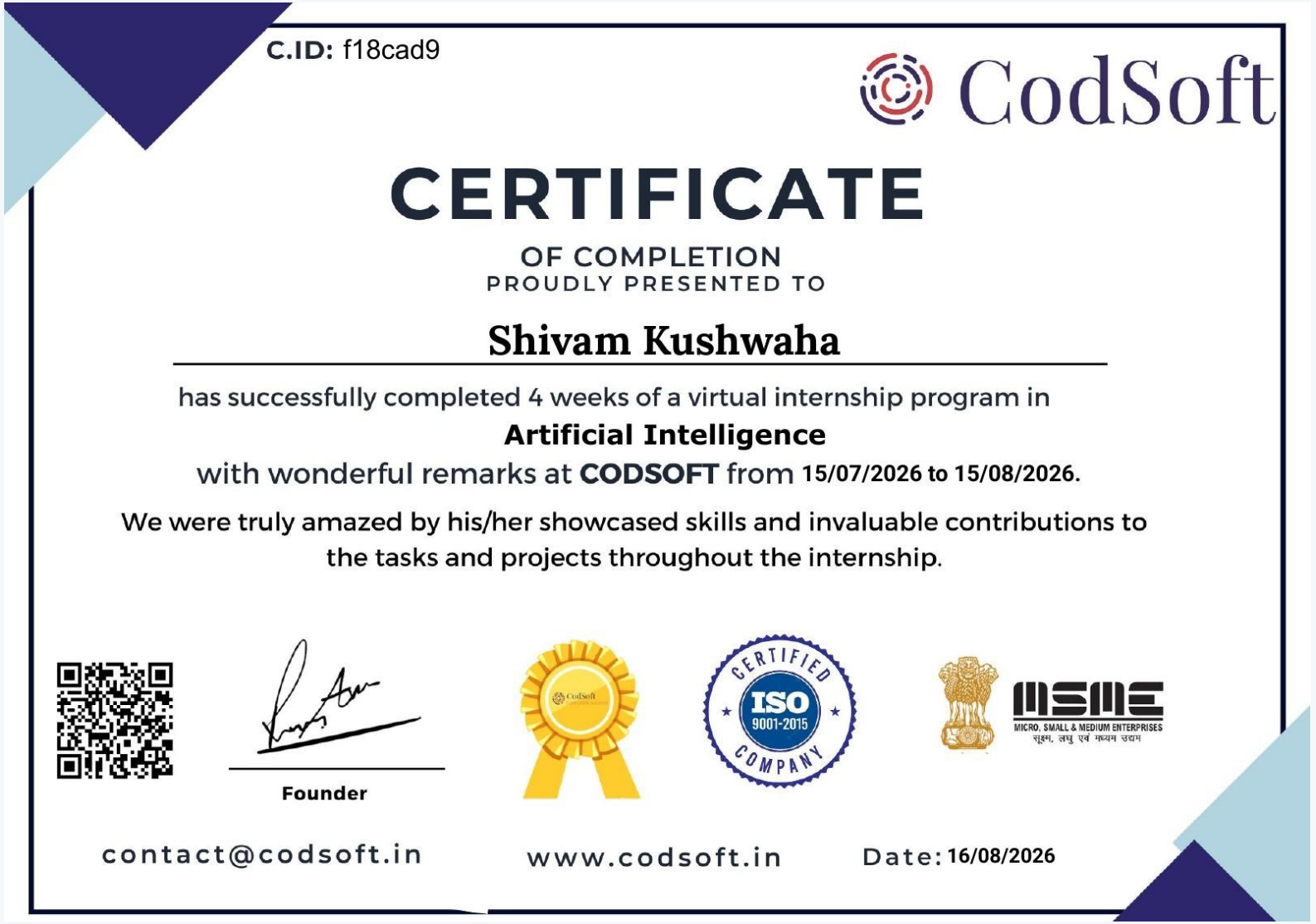
Git & GitHub

Version control and project tracking

Key Learning Outcomes

- ✓ Built a working rule-based chatbot capable of holding a basic conversation
- ✓ Developed an unbeatable Tic-Tac-Toe AI using Minimax with Alpha-Beta Pruning
- ✓ Created a functional recommendation system using filtering techniques
- ✓ Strengthened core AI concepts: NLP basics, adversarial search & personalization
- ✓ Gained hands-on experience translating AI theory into working Python code

Certificate of Completion



Thank You

Questions & Discussion

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